

ii) **Itemized Rating Scale**

In this scale, respondents are provided with a scale having numbers/ descriptions associated with each category. The respondents are asked to select the best fitting category with the object. The commonly used itemised rating scale are:

- a) Likert Type Scale (Summated Scale)
- b) Semantic Differential Scale
- c) Stapel Scale

a) **Likert Type Scale (Summated Scale)**

Likert (1932) proposed a simple and straight forward method for scaling attitudes that is most frequently used today. This scale is also known as summated rating scale. Summated scales consist of a number of statements which express either a favourable or unfavourable attitude towards the given object to which the respondent is asked to respond to each of the statements in terms of several degrees of agreement or disagreement. Let us understand this method with an example.

Against the following statements relating to psychological well-being, please tick mark any one of the following options shown against the statement

Sl. No.	Statements	Strongly Agree	Agree	Undecided	Disagree	Strongly Disagree
		5	4	3	2	1
1	I always live in present and do not worry about future.					
2	Many people think without purpose in life but I am not such type of persons.					
3	I generally feel sense of perfection during the moments I live in.					
4	I am very well in managing and discharging most of my responsibilities.					
5	I give importance to the innovative ideas and experiences that challenge my thought about me and the world.					

Depending on the wording of an individual item, an extreme answer of strongly agree or strongly disagree will indicate the most favorable response on the underlying attitude measured by the questionnaire. This scale is relatively easy and quick to compute. The data in this scale is of interval scale.

Some of the important advantages of this scale are: (i) it is relatively easy to construct as it can be performed without a panel of judges, (ii) it is considered more reliable because respondents answer each statement included in the

instrument, (iii) it can easily be used in respondent centred and stimulus centred studies, (iv) this takes less time to construct. However, there are several limitations of this type of scale i.e. (i) with this scale, we can simply examine whether respondents are more or less favourable to a topic but it is difficult to tell how much more or less they are, (ii) This does not rise more than to an ordinal scale structure, (iii) the total score of an individual respondent has little clear meaning since a given total score can be secured by a variety of answer patterns.

b) Semantic Differential Scale

This scale was developed by Osgood, Suci & Tannenbaum (1957). Semantic differential scale includes a seven point scale in comparison to Likert's five point scale. The semantic differential scale is based on the proposition that an object can have several implied or suggestive meanings to an expressed opinion. The semantic differential scale can be scored on either -3 to +3 or 1 to 7. This scale can also provide interesting comparison between products organizations and so on. The results of this scale are further analysed by the factor analysis.

Example: A researcher has developed an item seeking your perceptions regarding magazine. Please mark on each line that best indicates your perception. Please be sure to mark every scale and do not omit any scale.

India Today Magazine is

	+3	+2	+1	0	-1	-2	-3	
Easy to read	___	___	___	___	___	___	___	Hard to read
Modern	___	___	___	___	___	___	___	Old fashion
Rational	___	___	___	___	___	___	___	Emotional
Unreliable	___	___	___	___	___	___	___	Reliable
Worthless	___	___	___	___	___	___	___	Valuable
Unorganized	___	___	___	___	___	___	___	Organized
Orthodox	___	___	___	___	___	___	___	Liberal
Vain	___	___	___	___	___	___	___	Modest

Semantic differential scale provides a very convenient and quick way of gathering impressions on one or more than one concept. The data generated from this scale can be considered as numeric in some cases, and can be summed to arrive total scores. It must define a single dimension and each pair must be bipolar opposites labeling the extremes.

Like Likert Type Scale, this scale has also several advantages: (i) it is an efficient and easy way to secure attitude from a large sample, (ii) the total set of responses (both direction) gives a comprehensive picture of the meaning of an object as well as a measure of the subject of the rating, (iii) it is a standardised technique that can be easily repeated.

However, this technique escapes many of the problems of response distortion found with more direct methods.

c) Stapel Scale

Stapel scales are named after John stapel who developed these scales. Stapel scale consists of a single criterion in the centre with 10 categories numbered from -5 to +5 without a neutral point (zero). The scale is usually presented vertically. Negative rating indicates that the respondent inaccurately describes the object and positive rating indicates that the respondent describes the object accurately.

Example

Rate the Departmental store by marking (✓) on the following factors. +5 indicate that the factor is most accurate for you and -5 indicate that the factor is most inaccurate for you.

+5		-5	
+4		-4	
+3		-3	
+2		-2	
+1		-1	
Services		Products	
-1		+1	
-2		+2	
-3		+3	
-4		+4	
-5		+5	

In this scale, data generated is interval data. In this method, data can be collected through telephonic interview. The data obtained from Stapel scale can be analysed in the same way as semantic differential scale.

iii) Category Scale

In Category scaling method, objects are grouped into a predetermined number of categories on the basis of their perceived strength along certain dimension. This scale is a dichotomous scale. This method is useful for socio demographic questions. The response is this category we get, typically 'Yes' or 'No' type. Data in this category is either nominal or ordinal. Under this category scale, instructions and response tasks are quick and simple, and many options can be included.

Example

A) Single Category Scale

1) Do you own a house?

Yes	No

2) Do you own a car?

Yes	No

3) Do you own a Mobile phone?

Yes	No

b) **Multiple Category**

1) Please indicate your income by marking (✓) in the income group you fall

5000 - 10000	
10000 - 15000	
15000 - 20000	
20000 - 25000	

iv) **Cumulative Scale (Guttman Scale)**

Like other scales, it consists of series of statements to which a respondent expresses his agreement or disagreement. The statements are in a form of cumulative series i.e. in a way, an individual who replies favourably to say item no. 3 also replies favourably to item no. 2 and 1 and so on. The individual's score is worked out by counting the number of points concerning the number of statements he answers favourably. Knowing the total score, we can estimate as to how a respondent has answered individual statements constituting cumulative scales.

Let us understand with the illustration of an **example**.

Here is an example of a Guttman scale - the Bogardus Social Distance Scale:

(Least extreme)

- 1) Are you willing to permit immigrants to live in your country?
- 2) Are you willing to permit immigrants to live in your community?
- 3) Are you willing to permit immigrants to live in your neighbourhood?
- 4) Are you willing to permit immigrants to live next door to you?
- 5) Would you permit your child to marry to an immigrant?

(Most extreme)

Cumulative scale (scalogram analysis) like any other scaling technique has several advantages as well as limitations. The advantages are – It assures that only a single dimension of attitude is measured, (ii) Researcher's subjective judgement is not allowed to creep in the development of scale since the scale is determined by the replies of respondents.

Its main limitation is that (i) In practice perfect cumulative or unidimensional scale are very rarely found and approximation is used, (ii) Its developmental procedure is cumbersome in comparison to other scaling methods.

Check Your Progress 2

1) What is the distinction between comparative measuring scale and non comparative measuring scale?

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2) What do you mean by Likert Scale? Give its two examples.

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3) What are the sources of measurement errors?

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8.8 LET US SUM UP

Operationalisation of concepts for research purpose need measurement of various attributes of the people, the characteristics of objects or phenomenon. Measurement is a process to assign numbers to the characteristics according to certain rules. Measurement scales are of four types – nominal, ordinal, interval and ratio. A number of concerns arise in the process of measurement. Important among them are: the scale measurement i.e. the validity, reliability and practicality, number of indicators for measurement etc. Scaling techniques are broadly of two types – comparative, and non-comparative. Within comparative category, paired, rank order and constant sum are important techniques. Within non-comparative techniques, four sub-categories i.e. continuous rating scale, itemized rating scale, category scale and cumulative scale have been covered in this unit. All these scaling techniques have their relative advantages and limitations and are used depending upon the goal and the nature of characteristics of objects/phenomenon to be measured.

8.9 KEY WORDS

- Measurement** : The assignment of numerals to objects or events according to rules.
- Nominal** : Numbers are used to name identity or classify persons, objects, groups or gender.

Ordinal Scale	: Scale with order is called ordinal scale.
Interval Scale	: In this scale numerically equal distance on the scale indicate equal distances in the attributes of the object being measured.
Ratio Scale	: A conceptually meaningful zero point in which there is a total absence of the characteristics being measured.
Validity	: The extent to which an instrument or scale tests or measures what the measurement or scale i.e. consistent or dependent.
Reliability	: The degree to which the measurement or scale i.e. consistent or dependent.
Practicality	: Measure should be economical convenient and interpretable.
Errors in Measurement	: Discrepancies between the obtained scores and the corresponding scores.
Scaling	: The way of generation of a continuum upon which measured objects are located.
Comparative Scaling	The comparison of objects directly with one another.
Paired Comparison	: A binary choice, respondents choose the objects in each pair that has the greater magnitude on the choice dimension.
Rank Order	: Respondents are provided with various objects and asked to rank the list of objects.
Constant Sum	: Respondents are asked to allocate a constant sum of units to a specific set of objects with respect to predefined criterion.
Non Comparative Scaling	: Scaling of object independently to some specify standard.
Continuous Rating Scale	Respondents indicate their rating by marking at appropriate position on a line.
Itemised Rating Scale	Respondents are asked to select the best fitting category with the object.
Likert Scale	: Respondents are asked to select the best fitting category with the object.
Semantic Differential Scale	Seven point scale based on the proposition that an object can have several implied or suggestive meanings to an expressed opinion.
Stapel Scale	: A single criterion in the centre with 10 categories numbered from -5 to +5 without a neutral point.
Category Scale	: Objects are grouped in to a predetermined number of categories on the basis of their perceived strength along certain dimension.

8.10 SOME USEFUL BOOKS/PAPERS

Stevens S.S. (1946): *On Theory of Scales of Measurement, Science New Series*, Vol. 103, No. 2684, pp 77-680.

Kothari C.R. (2008): *Research Methodology Methods and Techniques*, New Age International Publishers. Chapter 5 page 69 to 82.

Babbie Earl, (2010): *The Practice of Social Research*, 12th Edition, WodsworthCengage Learning, CA, USA.

Malhotra, N.K. and S. Dash (2009): *Marketing Research, An Applied Orientation*, Pearson Education, New Delhi.

Singh, A.K.: *Tests Measurements and Research Method in Behavioural Sciences*, Bharti Bhawan Publishers and Distributors, Patna.

Gregory R.J. (2006): *Psychological Testing History, Principles and Applications*, Pearson Education, New Delhi, India.

Michael S. Lewis-Beck (ed) (2004): *The Sage Encyclopedia of Social Science Research Methods*, Vol. 2, Page no. 161 to 165, Sage Publications, New Delhi, India.

8.11 ANSWERS OR HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) See Section 8.2
- 2) See Section 8.4
- 3) See Section 8.5
- 4) See Section 8.3

Check Your Progress 2

- 1) See Sub-section 8.7.1 and 8.7.2
- 2) See Sub-section 8.7.2 under the subhead Itemised Rating Scale
- 3) See Section 8.6.